Fiscal Year:	FY 2018	**Task Last Updated:**	FY 12/12/2017
PI Name:	Tash, Joseph S Ph.D.		
Project Title:	Female Reproductive Health: Space Flight Induced Ovarian and Estrogen Signaling Dysfunction, Adaptation, and Recovery		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	02/01/2015	**End Date:**	01/24/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	Tissue Sharing NOTE: End date changed to 1/24/2018 due to change in PI (Ed., 1/31/18)		
Key Personnel Changes/Previous PI:	NOTE (January 2018): Project continues as "Female Reproductive Health: Space Flight Induced Ovarian and Estrogen Signaling Dysfunction, Adaptation, and Recovery (PI: Christenson)" with the same grant number NNX15AB48G due to retirement of Principal Investigator Joseph Tash, Ph.D.		
COI Name (Institution):	Christenson, Lane Ph.D. (University of Kansas Medical Center Research Institute) Ronca, April Ph.D. (NASA Ames Research Center) Alwood, Joshua Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	NNX15AB48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our studies conducted on three space shuttle missions (STS-131, STS-133, and STS-135) represented the first systematic studies of ovarian function ever in normal adult cycling mice in space. Our results demonstrated disruptions in both follicular development and the corpus lutea formation, both of which would render these females infertile. This dramatic ovarian dysfunction occurred in the absence of increased stress biomarkers and was associated with a pronounced loss of estrogen receptor expression in the uterus. In view of NASA's plans to conduct multigenerational studies on rodents in long-term space flight, these observations raise important fundamental questions on maintenance of female reproductive health that need to be addressed prior to conducting complex multigenerational studies in mammals under space flight conditions. As a result of our previous findings, specific important knowledge gaps now exist regarding the impact of space flight on female reproductive health. The central hypothesis that we will test is that space flight elicits direct and/or indirect effects at the ovarian level to cause infertility in adult female mice. This grant application combines the expertise of Dr. Tash in fertility regulation and reproductive physiology, Dr. Christenson in ovarian physiology, and Dr. Ronca in pregnancy and early development, to address this central hypothesis. Additionally, these studies will address an important secondary hypothesis that the loss in uterine estrogen receptor expression will also occur in bone and this may exacerbate the observed bone mass loss during space flight (Dr. Alwood as collaborator). To address these hypotheses the following specific aims are proposed: Aim 1. Determine whether short-term (~10 d) space flight-disrupted ovarian function is due to a direct effect at the ovarian level. The working hypothesis, based on our preliminary data, is that disrupted ovarian function is due to a failure of the periovulatory follicle to ovulate and form a functional corpus luteum. To test this hypothesis, mice will undergo a timed follicular stimulation protocol followed by evaluation of established temporally expressed ovarian markers, ovulation, and luteal development. This study will for the first time determine if space induced loss of ovarian function is due to an intrinsic ovarian defect or is a problem with hypothalamic/pituitary support. Aim 2. Determine if adult female mice recover ovarian function during extended (~60 d) flight time. Currently, it is unknown whether females will recover fertility following long-term (i.e., ~60 d) acclimation to space flight, or will they remain infertile or exhibit more pronounced defects in ovarian/uterine function. To address this question, mice will undergo a similar follicular stimulation and ovulatory protocol in flight but after ~60 d, as described in Aim 1, and fertility and tissue biomarkers will be evaluated. Aim 3. Determine if adult female mice exposed to space flight are fertile upon return to Earth. The working hypothesis for this aim is that upon return to Earth, fertility will return. To test this hypothesis, flight mice will be exposed to male mice during a recovery period upon return to Earth, and regain of mating reproductive behavior, estrous cyclicity, early embryo implantation, and health and development of F1 offspring will be evaluated. These studies will provide the first ever examination of long-term space exposure on the female reproductive behavior/physiology, and germ-line stability. In all 3 specific aims, collection of uterine and bone (femur) tissues will allow us to address our secondary hypothesis by determining long-term space flight effects on estrogen receptor expression. Ultimately, findings of the proposed studies will provide critical foundational data to support multigenerational mammalian studies in space, and importantly, aid in our understanding of the impact space flight has on female astronaut health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fertility is a general measure of fitness of a species on Earth. Disruptions to fertility have both short and long term effects on the health on the individual or the ability of a species to maintain a critical numbers, respectively. Earth species have adapted to changes in environment over millennia and it is these adaptations that allow species to survive and flourish on Earth. The reproductive system of both female and male mammals are well tuned to sense environmental changes. These environmental responses can then either increase or decrease fertility appropriately. The effects of weightlessness on female ovarian function we currently are studying are critical as we consider longer term flights and the overall health of these individuals upon return to Earth. The fact that ovarian estrogen plays a key metabolic role in almost all tissues of the body also highlights the importance of understanding how female fertility is influenced by exposure to space. The Earth benefits of this research may yield new yet undiscovered mechanisms controlling ovarian function, steroidogenesis, and female fertility. Our research has led us to adapt an approach to monitor estrous cyclicity in rodents; we believe this methodology has the potential to impact all female based NASA flight research projects due to our ability to accurately determine the stage of the estrous cycle and thus predict ovarian estrogen levels.
Task Progress:	Ovarian estrogen impacts the normal homeostatic and metabolic processes of all tissues and organ systems within the body: particularly, but not limited to canonical space-flight impacted systems: bone, muscle, immune, wound repair, and cardiovascular. Effects of space flight on the ovarian estrogen production are therefore critical to our understanding of all space flight experiments using female mice, the current paradigm being used on the International Space Station (ISS). Recently, we demonstrated that vaginal wall histology could be used to determine the stage of the estrous cycle in female mice at the time of sacrifice in space. Moreover, this robust technique was completed following two post-flight freeze/thaw procedures of the carcasses (RR1 experiment). Thus, this technique represents a viable mechanism to determine the estrous cycle status of the female at the time of sacrifice and can be completed in a manner that does not impact primary experimental objectives. We propose that vaginal wall histology become a standard procedure completed on all mice sacrificed in space and that the individual estrous status of each animal be shared with all investigators. While evidence of estrous cyclicity was present in long-term (33 day) RR1 mice, fertility of female mice exposed to weightlessness remains unknown. In preparation for an upcoming funded NASA flight investigating the effects of long duration space flight on female fertility, we have refined our experimental design to minimize crew time and to accommodate the duration of Dragon capsule berth. These refinements maintain all our proposed primary and secondary experimental objectives. Briefly, in order to evaluate fertility, we will super ovulate mice using standard procedures (PMSG ± hCG), followed by collection of reproductive tract after follicular stimulation alone (PMSG) or following ovulation (hCG). Ovarian folliculogenesis and ovulation rate will be determined in fixed tissues following return in order to determine fertility. Ovarian and uterine tissues will also be evaluated by hormonal and gene expression profiling using quantitative approaches (radioimmunoassays, western blots, digital droplet PCR). Comparisons will be made to contemporary vivarium and Rodent Research Hardware –Transporter and Habitat housed animals maintained on Earth. Hindlimb unloading (HU) is a rodent model system used to simulate weightlessness experienced in space. However, some effects of this approach on rodent physiology are under-studied, specifically the effects on ovarian estrogen production, which drives the estrous cycle. To resolve this deficiency, we conducted a ground-based validation study

	using the HU model, while monitoring estrous cycles in 16-weeks-old female C57BL6 mice. Animals were exposed to HU for 12 days following a 3 day HU cage acclimation period, and estrous cycling was analyzed in HU animals (n=22), normally loaded HU Cage Pair-Fed controls (CPF; n=22), and Vivarium controls fed ad libitum (VIV; n=10). Pair feeding was used to control for potential nutritional deficits on ovarian function. Vaginal cells were sampled daily in all mice via saline lavage. Cells were dried and stained with crystal violet, and the smears evaluated using established vaginal cytology techniques by two individuals blinded to the animal treatment group. Estrous cyclicity was disrupted in nearly all HU and CPF mice, while those maintained in VIV had an average normal cycle length of 4.8 ± 0.5 days, with all stages in the cycle visibly observed. CPF and HU animals arrested in the diestrous phase, which precedes the pre-ovulatory estrogen surge. Additionally, infection-like symptoms characterized by vaginal discharge and swelling arose in several HU animals, which we suspect was due to an inability of these mice to properly groom themselves, and/or due to the change in the gravity vector relative to the vaginal opening, which prevented drainage of the lavage solution. Pair-feeding resulted in similar weight gains of HU and CPF (1.5% vs 3.0%, respectively). The current results indicate that pair-feeding controlled weight gain and that the HU cage alone influenced estrous cyclicity. Thus, longer acclimation needs to be tested to determine if and when normal estrous cycling resumes in non-loaded mice in HU cages prior to HU testing. Future studies might also examine whether modifications to the vaginal lavage procedure might prevent the onset of the infection-like symptoms, and allow estrous cyclicity to be measured in this model system. NOTE (January 2018): Project continues as "Female Reproductive Health: Space Flight Induced Ovarian and Estrogen Signaling Dysfunction, Adaptation, and Recovery (PI: Christenson)" with the same grant number NNX15AB48G due to retirement of Principal Investigator Joseph Tash, Ph.D. See that project for future reporting.
Bibliography Type:	Description: (Last Updated: 01/24/2022)
Abstracts for Journals and Proceedings	Christenson LK, Hong X, Alwood JS, Ronca AE, Tash JS. "Development of a novel space flight plan to monitor female mice fertility using minimal astronaut flight time." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. Oral presentation. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Moyer EL, Talyansky Y, Scott RT, Tash JS, Christenson LK, Alwood JS, Ronca AE. "Estrous Cyclicity of Mice During Simulated Weightlessness." Presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Christenson LK, Hong X, Choi S, Ronca AE, Alwood JS, Tash JS. "Evaluation of estrous cycle stage in mice following long-term space flight using vaginal histology." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Articles in Peer-reviewed Journals	Hong X, Ratri A, Choi SY, Tash JS, Ronca AE, Alwood JS, Christenson LK. "Effects of spaceflight aboard the International Space Station on mouse estrous cycle and ovarian gene expression." NPJ Microgravity. 2021 Mar 12;7(1):11. https://doi.org/10.1038/s41526-021-00139-7 ; PMID: 33712627; PMCID: PMC7954810 , Mar-2021